

# Investigating Spectral Attenuation of Visible Light with Depth in Coastal Waters

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**Abstract—** This paper presents the design and deployment of an Autonomous Underwater Vehicle (AUV) built for E80: Experimental Engineering at Harvey Mudd College. The AUV was designed with the goal of investigating how visible light intensity and spectral composition attenuate with depth in coastal waters that harbor kelp ecosystems. The rapid decline of kelp forests along the Southern California coast motivated the need to analyze the underwater light environment that supports kelp growth and photosynthesis. The AUV integrated an MPX5700 pressure sensor, a Murata NXFT15WB473FA2B150 NTC thermistor, and a TCS34725 RGB color sensor. The AUV was deployed approximately 30 meters offshore Dana Point, California via kayak. After two successful dives collecting depth and temperature data to a depth of 3 meters using the pressure sensor and thermistor, saltwater entered the electronics housing, short-circuiting the protoboard. A third, tether-controlled dive collected RGB sensor data to measure spectral light intensity to a depth of 3 meters. Results showed that more than 80% of surface light was lost at a depth of 3m, with red wavelengths attenuating most rapidly. Only 8.3% of surface red light remained at 3m, compared to a 9.7% for green and 10.1% for blue, consistent with Beer-Lambert predictions and the known absorption spectrum of water. The thermistor malfunctioned as it recorded a fixed temperature of 23.7°C throughout the dive, despite a recorded ground truth surface measurement of 18.7°C, and no reliable temperature data was collected. From the results, it was concluded that kelp at Dana Point is able to survive at depths greater than 3m, as the minimum for photosynthesis, in accordance with Photosynthetically Active Radiation (PAR), was not reached at the maximum dive depth, as certain kelp species require only 1% of surface light.

## I. INTRODUCTION

The goal of this project is to investigate how visible light intensity and spectral composition attenuate with depth in coastal waters at Dana Point, California. Kelp forests are an essential part of the coastal ecosystems, providing habitat and a food source for marine species [1]. Kelp forests are dependent on sunlight for photosynthesis and growth. Recently, kelp forests along the California coast have experienced a rapid large-scale decline, motivating the goal of understanding the underwater light environment as depth changed and how it is currently supporting kelp growth [2].

The Beer-Lambert Law states that light intensity in water decreases exponentially with depth [3]:

$$I(z) = I_0 * e^{-\alpha z} \quad (1)$$

where  $I(z)$  is light intensity at a depth  $z$ ,  $I_0$  is light intensity at surface water level, and  $\alpha$  is the attenuation coefficient. It is known that in water, red wavelengths are absorbed most strongly, as they attenuate the quickest. Blue wavelengths demonstrate intermediate penetration in coastal waters due to turbidity and sediments, whilst green wavelengths are able to penetrate to greater depths, as seen in Figure 1 [4].

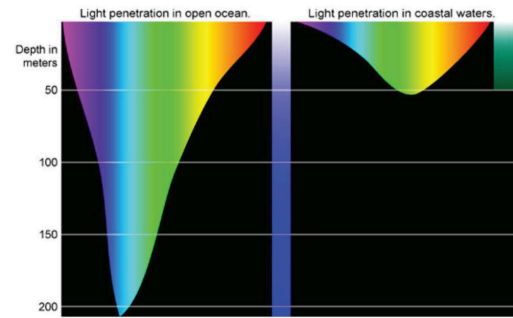


Figure 1. Light absorption spectrum as depth increases in open ocean (left) and coastal waters (right)

This spectral attenuation (RGB) is directly linked to kelp growth, which requires only 1% of surface Photosynthetically Active Radiation (PAR), which is wavelengths 400-700nm, for photosynthesis [6]. By measuring how red, green, and blue light attenuate with depth, the maximum depth and depth range at which kelp photosynthesis is possible at Dana Point can be estimated. Additionally, temperature is a factor in kelp growth and an analysis of water temperature is beneficial in understanding the growth environment of kelp [11].

With this goal in mind, an AUV was designed integrating a pressure sensor, thermistor, and a RGB color sensor. It was expected that total light intensity would decrease exponentially as depth increased and that red light would attenuate the fastest, consistent with the light absorption spectrum seen in Figure 1. This report describes the design and deployment process of the AUV in order to capture light attenuation, depth, and temperature data. It also describes the sensor design, calibration, deployment protocol, with discussion of results, and recommendations for future work.

## II. EXPERIMENTAL SETUP

### A. Engineering Goal

The team aimed to explore how visible light attenuated spectrally (RGB) with depth near kelp ecosystems present in

the coastal waters near Dana Point, California. The primary focus was a spectral light attenuation visualization relating red, green, and blue light transmission to depth, in order to estimate the depth limit for kelp photosynthesis to occur. Additionally, the AUV aimed to use a pressure sensor to estimate dive depth and record water temperature with depth.

### B. Sensor Selection

Three sensors were selected to measure depth, temperature and spectral light intensity. The MPX5700 pressure sensor was selected as the depth measurement tool, as it has a linear relationship between water pressure and output voltage that allows for reliable underwater depth estimation [7]. The Murata NXFT15WB473FA2B150 NTC thermistor was chosen to measure temperature due to its sensitivity (B constant of 4050K) and an operating range suitable for the expected 11-22 °C at Dana Point [8]. The TCS34725 RGB Color Sensor was selected in order to measure red, green, blue and clear light intensity though four separate filtered photodiodes with an internal analog-to-digital converter (ADC), allowing for the collection of a digital spectral measurement without the need for external circuitry [9]. All sensors were controlled with a Teensy 4.1 microcontroller, with both the pressure sensor and thermistor using the Teensy analog interface (pins A0 and A1, respectively), while the RGB sensor communicated with the Teensy through I2C.

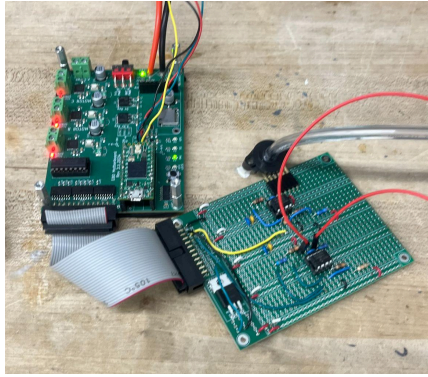


Figure 2. Motherboard (left) with RGB wiring and Protoboard (right) with Pressure Sensor, and Thermistor circuitry wiring

### C. Pressure Sensor Circuit Design

The MPX5700 pressure sensor was used to estimate depth over a range of 0-6 m. From the sensor datasheet, the output voltage of the sensor,  $V_{out(sensor)}$ , is related to pressure  $P$  (kPa), by the equation seen in Figure 3 [7]:

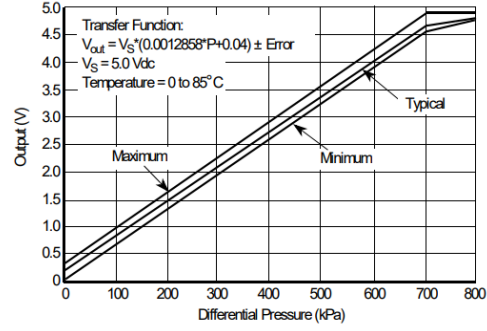


Figure 3. Output sensor voltage related to pressure

where  $V_s = 5V$  and pressure (kPa) at a depth  $h$  (m) is:

$$P(h) = P_{atm} + \frac{\rho gh}{1000} \quad (2)$$

Using standard atmospheric pressure ( $P_{atm} = 101.325 \text{ kPa}$ ) and equation 2, at 6m, the expected pressure is  $P = 160.185 \text{ kPa}$ . With known pressures and using the transfer function from Figure 3, the sensor voltage output ranges from 0.851 V at the surface (0 m) to 1.2310 V at a depth of 6 m. Since the Teensy's safe analog input range is 0.3-3.3 V, an inverting offset amplifier circuit, seen in Figure 4, was designed using an MCP6002 op-amp to scale and offset the sensor output to an expected range of 0.56 - 3.01 V [10].

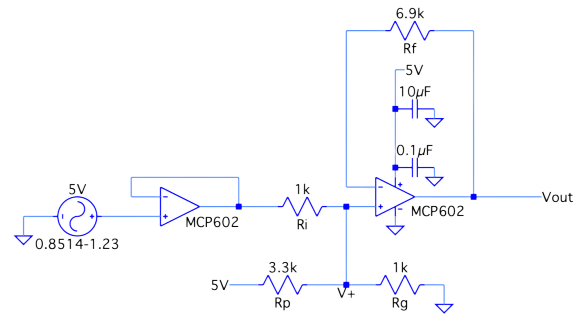


Figure 4. Pressure Sensor (MPX5700) Circuit Schematic with unity gain buffer B (left) and inverting amplifier A (right)

As seen in Figure 4, the initial op-amp stage (input B) was designed as a unity gain buffer on the non-inverting input (+ input A) in order to prevent signal loading that may come from the sensor. Terminals A of the op-amp were used for the inverting amplifier stage, in which the sensor and voltage divider were fed to the positive, non-inverting input, and the inverting stage occurred in the negative input terminal. With a desired range of 0.5-3.1 V, the inverting amplifier stage set a gain of 6.87 using resistor values  $R_i = 1 \text{ k}\Omega$  and

$R_f = 6.9 \text{ k}\Omega$ . With this gain, the voltage divider offset the signal to  $V_+ = 1.14 \text{ V}$  from the  $5 \text{ V}$  voltage supply using resistors  $R_g = 1 \text{ k}\Omega$  and  $R_{p1} = 3.3 \text{ k}\Omega$ . The resulting inverted signal will output an expected voltage range of  $3.01\text{-}0.56 \text{ V}$  at  $0\text{-}6 \text{ m}$ , respectively, that will safely be fed into Teensy analog pin A0.

The MPX5700 has a response time of approximately  $1 \text{ ms}$  [7], which is negligible relative to Teensy's sample rate of  $10 \text{ Hz}$  ( $100 \text{ ms}$ ).

#### D. Thermistor Circuit Design

An inverting amplifier circuit was constructed using a Murata NXFT15WB473FA2B150 NTC thermistor to measure water temperature. The resistance of the thermistor,  $R_T$ , varies with absolute temperature  $T$  (K), according to the B-parameter equation]:

$$R_T = R_0 * \exp(B * \frac{1}{T} - \frac{1}{T_0}) \quad (3)$$

where  $R_0 = 47 \text{ k}\Omega$ ,  $T_0 = 298.15 \text{ K}$ , and  $B = 4050 \text{ K}$  [8]. With the deployment temperature range of  $11\text{-}22^\circ\text{C}$ , a resistance range of  $R_T = 90 - 55.71 \text{ k}\Omega$  is expected at its respective temperatures. A voltage range of  $1.715\text{-}2.288 \text{ V}$  is expected to come from the voltage divider output at the deployment temperature range. To satisfy the Teensy's safe operating range of  $0.3\text{-}3.3 \text{ V}$ , the input was scaled and shifted using an inverting offset amplifier with a MCP6002 op-amp. With a gain of  $4.71$  from resistors  $R_n = 10 \text{ k}\Omega$  and  $R_f = 47 \text{ k}\Omega$  and offset resistors  $R_p = 15 \text{ k}\Omega$  and  $R_g = 10 \text{ k}\Omega$ , the voltage was safely fed into Teensy analog pin A1, As seen in Figure 5.

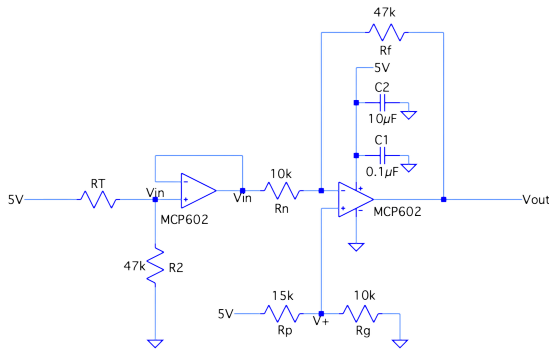


Figure 5. Thermistor(Murata NXFT15WB473FA2B150) Circuit Schematic

The thermistor's thermal time constant in water is  $4 \text{ seconds}$  [8]. At the Teensy's sample rate of  $10 \text{ Hz}$  ( $100 \text{ ms}$  samples), the sensor is sampled at a rate higher than the

time constant, meaning that although the thermistor has not settled, the Teensy still records the temperature changes.

#### E. RGB Color Sensor Circuit Design

The TCS34625 RGB color sensor measures the light intensity through four separate photodiodes filtered to red, green, blue, and clear light. Clear light is essentially an unfiltered light that can detect a wider range of wavelengths. All photodiodes wavelengths can be seen in Figure 6 below [9]. From the Figure, red has peak sensitivity at  $\sim 615 \text{ nm}$ , with a range of  $\sim 580\text{-}600 \text{ nm}$ , green has a peak at  $\sim 525 \text{ nm}$ , range of  $\sim 380\text{-}700 \text{ nm}$ , blue has a peak at  $\sim 465 \text{ nm}$ , range of  $\sim 380\text{-}800 \text{ nm}$ , and clear has a broad range of  $\sim 380\text{-}700 \text{ nm}$ . All light sensitivity seems to converge at wavelengths  $\sim 830 \text{ nm}$ .

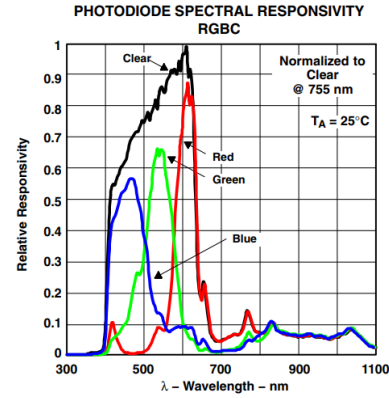


Figure 6. RGB sensor's photodiode light sensitivity wavelength range

Each photodiode generates a current proportional to the irradiance detected, in which a 16-bit ADC converts it into digital count values [9]. Because the ADC response and irradiance is linear, the ratio of counts at a depth to counts at the surface is able to provide a spectral transmission ratio. Essentially, the Beer-Lambert equation can be computed throughout the dive using the counts.

The sensor operates over the voltage range of  $2.7\text{-}3.6 \text{ V}$  and is powered directly from the Teensy's  $3.3 \text{ V}$  output. No external analog circuitry is needed for the sensor to function. Unlike the analog sensors, the TCS34725 RGB sensor communicates with the Teensy Via I2C, rather than ADC. To do this, the sensor's wiring was soldered into the motherboards' "testing" ports, which are also SDA on pin 18, and SCL on pin 19, as seen in Figure 7. Unfortunately, the Teensy could not read the sensor when wiring from the motherboard directly to the required pins was being tested. While the ports used on the motherboard are labeled "For Testing Only", they also feed directly to the required pins for the Teensy with the same functionality.

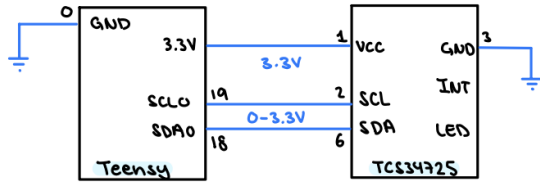


Figure 7. RGB Light Sensor (TCS34725) Circuit Schematic

The RGB sensor does not have a time constant, but rather an integration time of 24ms. During the Teensy 10Hz sampling rate, the RGB sensor collects data over 24ms each sample, averaging the light detected during this window. Although the window is smaller than the 100ms of the Teensy, light attenuation with depth was measured in a larger scale of time (minutes), meaning that the decision to use this sensor was practical, since a gradual decrease of light allows for better characterization of light attenuation with depth.

#### F. Sensor Modeling and Circuit Verification

To ensure the functionality of the sensors and to convert voltage outputs to relevant data, calibration curves and bench testing was conducted.

For the pressure sensor, the circuit output voltage was verified by submerging a tube connected to the sensor in a graduated cylinder filled with water at measured depths and recording output voltages with a multimeter and cross referencing with the Arduino Serial monitor. This was done multiple times to confirm the validity of the circuit and debug issues. Before deployment, the graduated cylinder was filled with saltwater from the deployment site the resulting calibration gave the equation:

$$V_{out} = -3.1 * h + 8.64 \quad (4)$$

where h is depth (m). Figure 8 demonstrates the curve with linear fit and confidence bounds.

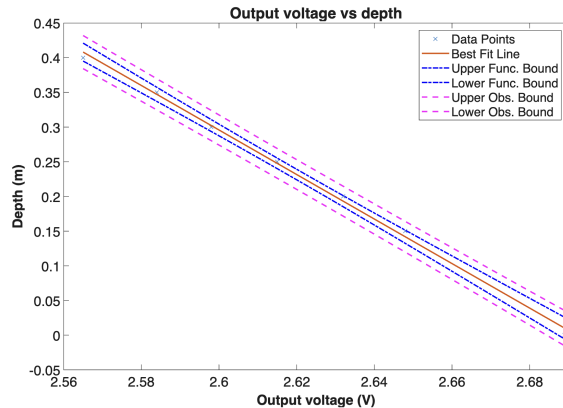


Figure 8. Pressure Sensor Calibration Curve (Seawater)

After calibration, the pressure sensor calculated its depth to be 0.5 m at surface level (0 m). During post-processing, the 0.5 m offset was corrected to ensure the data plotted was at the appropriate depths.

For the thermistor, MATLAB was used with the B-parameter equation (5) to verify the circuit produced expected output voltages at the extremes of the range 11-22°C. As for calibration, a Steinhart-Hart fit, seen in Figure 9, was produced by finding the resistance values at known temperatures with a multimeter.

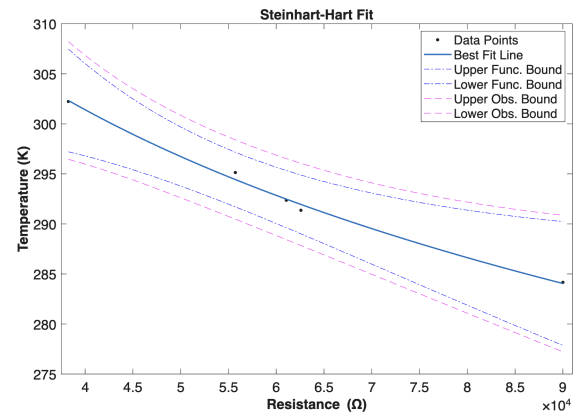


Figure 9. Steinhart-Hart fit of Thermistor (K vs  $\Omega$ )

As for the RGB sensor, calibration was conducted at Dana Point by recording two identical measurements facing the sun: one being the bare sensor and the other with the sensor inside the sealed clear waterproof box. The average counts were computed for each trial during a stationary window, as seen in Figure 10. Then, the correction factor for each channel was calculated as the ratio of bare sensor counts to in box counts and applied into all dive data. Table 1 demonstrates the results from calibration and this was done in order to correct any interference the clear acrylic filter of the box may have with the raw sensor data.

|                           | Red<br>(Counts) | Green<br>(Counts) | Blue<br>(Counts) |
|---------------------------|-----------------|-------------------|------------------|
| <b>Bare Sensor</b>        | R=9333          | G=10098           | B=10233          |
| <b>In-Box</b>             | R=8578          | G=9826            | B=10080          |
| <b>Correction Factors</b> | R=1.0881        | G=1.0277          | B=1.0152         |

Table 1. Calibration Results. Counts detected by the sensor and Correction Factor (ratio) for each color channel.

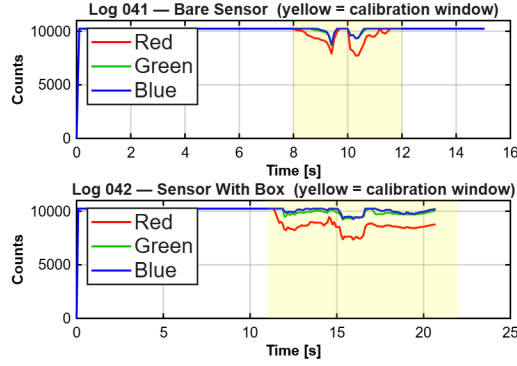


Figure 10. Trials for RGB sensor calibration

It should be noted that due to repeated motherboard and teensy short-circuits during the build and integration process, test trials for data collection in the tank room were limited to verifying buoyancy and motor functionality. Data collection only occurred during Dana Point deployment.

#### G. Mechanical Design

The AUV was initially designed as a compact, symmetric diving robot with a frame of 8" x 6" x 3" and estimated mass of 1090.7g, as seen in Figure 11. The symmetric design had the center of gravity at (3.375, 2.625, 2.166) in. in the x,y,z directions, respectively.

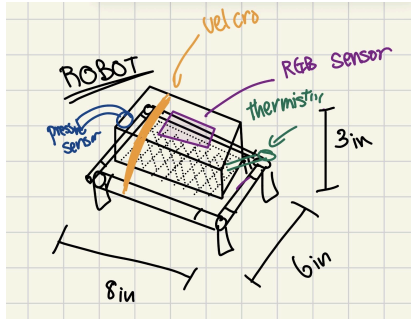


Figure 11: Original AUV design sketch

Prior to deployment, modifications were made to the design. The following remained the same: The main blue box housed the Teensy 4.1 microcontroller, the E80 motherboard [12], the protoboard with the pressure and thermistor circuits, a 33000 mAh battery, and ballast masses. A smaller box with a clear filtered enclosure was fixed on top of the box housing the RGB sensor; both housings were secured to the frame with velcro and zip ties.

Neutral buoyancy was calculated using a force balance when the AUV is in water:

$$m_{robot} = \rho_{water} V_{tot} \quad (5)$$

The modifications to our original design consisted of the following: The PVC frame was extended, an increase in width, as well as the arms being extended upwards, si. Thus, the final dimensions were 25 x 17 x 23 cm and the center of gravity was estimated to geometrically be located at (12.5, 8.5, 11.5) cm, with the same estimated mass as before, with removed ballast.. This new design allowed for pool noodles to be added and removed as necessary, given the buoyancy difference during Dana Point (saltwater) and testing (fresh water). A mesh was also added to secure the box to the extended frame and improve stability, with motors being attached through drilled holes of the horizontal PVC and secured to the main housing with zip ties, as seen in Figure 12a and 12b.

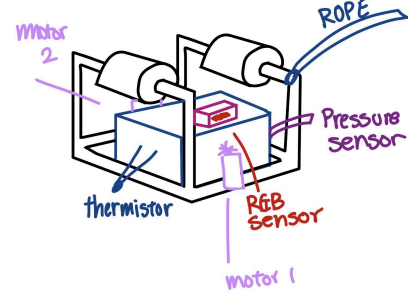


Figure 12a: Final AUV design sketch

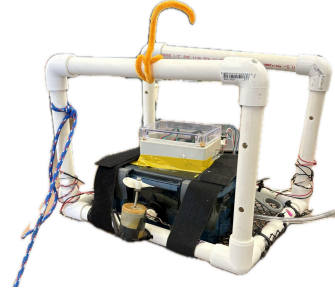


Figure 12b: Final AUV Robot

This design was still symmetric and optimal for a vertical autonomous dive. As seen in the figures above, the pressure sensor and thermistor were mounted on the sides of the electronics housing and waterproofed with epoxy-sealed penetrators. The RGB sensor was in a separate compartment with a penetrator connecting the RGB sensor to the main box. This design choice was to ensure the RGB sensor was facing upward in a clear filter for accurate light measurements. A waterproofing issue with the RGB housing was resolved by replacing the original connector penetrator with individual wires and a smaller epoxy bolt penetrator, as well as applying parafilm in possible gaps. Washers were added where the penetrators were located, to minimize risks of flooding.

The estimated total current draw was approximately 1634 mA, with a resulting UAV mission life of ~2 hours.

#### H. Experimental Procedure

During the building process, the team revised the intended depth of 12m to 6m after further investigation of the depth being closer to ~6m at the deployment site. Deployments were initially planned to be conducted at different times of the day to capture differences due to the sunlight angle, but flooding of the electronics housing after the initial dives prevented further data collection.

Before deployment, all sensor circuits were dry tested, validating the pressure and thermistor outputs. Waterproofing of all penetrators and security to the frame was confirmed, as well as attaching a tether. Code was uploaded to Teensy with a 2-minute startup delay to close the main electronics housing and check that it is properly attached to the frame.

Autonomous driving used proportional (P) control, in the vertical motor effort,  $uV$ , was computed each loop period as:

$$depth\_error = depth\_des - depth \quad (6)$$

$$uV = Kp * depth\_error, \text{ where } Kp = 150 \quad (7)$$

The robot paused at each depth waypoint for 10 seconds, with a depth margin of 0.05 m. Waypoints were set to 1 m, 1.75 m, and 3 m to test UAV path before intending to collect data at deeper waypoints until reaching 6 m. Throughout the dive, pressure, temperature, and RGB data was collected with a microSD card. After the final waypoint, the UAV would resurface without stopping.

As for code modifications, the DFRobot\_TCS34725 [13] Arduino library was integrated into the starter code in order to communicate with the RGB sensor with the Teensy via I2C. Within the code, a *SensorRGB.h* file was created, logging the four channel (RGBC) counts to the SD card while diving. Within the main *Integrated.Robot.ino*, *SensorRGB.h* is called, updating the counts after every 10 Hz. Additionally, normalization lines were set up to prevent a crash caused by division by zero in the case the clear channel reads 0.

Three dives were successfully conducted. Dive 1 was motor-assisted, with successful pressure and thermistor data collection. The robot seems to have skipped over the waypoints and it is likely due to the rapid descent compared to the small 5 cm depth margin. Dive 2 was conducted with the motors off to analyze what noise was seen in dive 1 due to the motors. However, saltwater flooded the main electronics housing, most likely due to a loose wire preventing a full

lid closer, short-circuiting the breadboard, with an unsuccessful data collection.

The motherboard survived the flooding, allowing the collection of RGB sensor data with depth. A third dive was conducted with the RGB sensor only, with a manual tether marked at 1m intervals, descent was controlled manually, pausing at 0m (surface), 1m, 2m, and 3m. The five occurred near noon (~12pm). RGB data was successfully collected. All data was post-processed in MATLAB with calibration and correction factors.

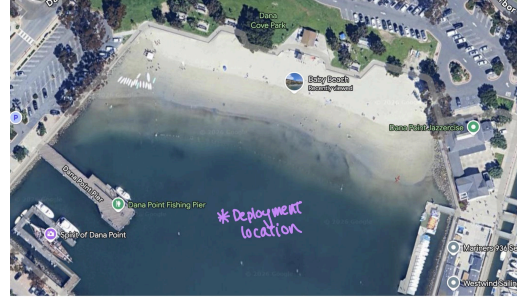


Figure 13: Deployment Location (purple) at Dana Point, CA, 30m offshore via kayak

Safety was maintained during deployment by keeping a tether on the AUV at all times, the kayaking team members wore life vests, and members remained close to the UAV during the dive.

### III. RESULTS AND ANALYSIS

#### A. Deployment Overview

The AUV was deployed at Dana Point, California, at approximately 30 m offshore via kayak on Saturday, April 25th, 2026. Three dives were conducted. Dive 1 collected pressure and thermistor data up to 3 m. Dive 2 intended to collect data with the motors turned off to prevent motor noise interference, but saltwater flooded the electronic housing, frying the protoboard and ending further autonomous data collection. Dive 3 collected RGB spectral attenuation data via a tether manually until a depth of 3 m to match the data collected from the other sensors. Refer to Section II. H. for additional information.



Figure 14: E80 Section 4, Team 45 with Final AUV before deployment

### B. Pressure Sensor Results

Before deployment, the pressure sensor was calibrated using a graduated cylinder filled with saltwater, resulting in Equation (4). After calibration, a 0.5 m surface offset was revealed, which was subtracted during post processing to produce the Depth vs Time profile seen in Figure 15. During Dive 1, the pressure sensor successfully tracked depth throughout the dive and P-control held depth at the 3 m final waypoint before surfacing, but did not stop at the waypoints, likely due to the small depth\_margin of 0.05 m.

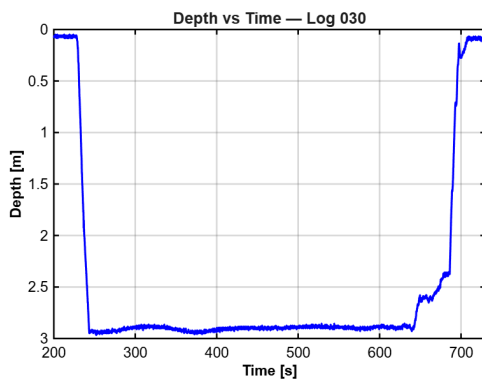


Figure 15: Depth vs. Time (Dive 1)

### C. Thermistor Results

The thermistor recorded a fixed temperature of  $23.7^{\circ}\text{C}$  throughout the dive with no changes with depth, as shown in Figures 16 and 17. A ground truth surface temperature of  $18.7^{\circ}\text{C}$  was established with a thermometer prior to the dive. Additionally, the circuit was tested beforehand and validated.

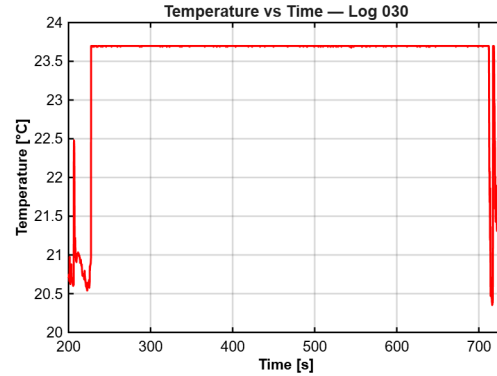


Figure 16: Temperature vs. Time (Dive 1)

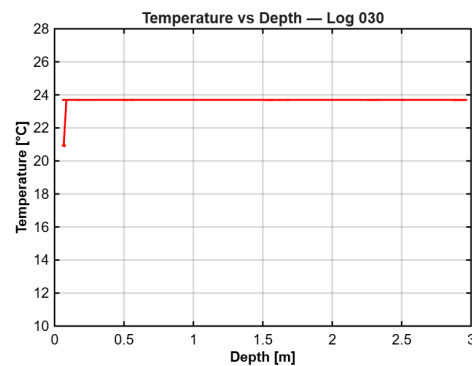


Figure 17: Temperature vs. Depth (Dive 1)

There were multiple theories that may explain the behavior seen in Dive 1. The fixed high temperature demonstrates behavior that would happen had it been an open circuit. Thus, three reasons may have caused the fixed  $23.7^{\circ}\text{C}$ . 1. As the robot was under stress from the water, the wire was pulled out of the protoboard (which is very unlikely as it was checked after the dive and seemed connected to the board). 2. The thermistor itself may have been damaged, as the previous day the thermistor circuit was successfully tested with a different thermistor, but was unable to be used due to no longer being waterproofed. Or 3. The waterproofing of the thermistor insulated the thermistor, preventing readings from the water. As a result, no reliable temperature data was collected.

### D. RGB Color Sensor Results

RGB data was collected during Dive 3. From the calibration, Box correction factors were applied to all raw counts before graph analysis. A spike visible in Figure 18, above the surface level was caused by the box lid being briefly reopened due to a looser wire, in which the sensor faced the

sun directly (at an angle), recording the elevated signal counts.

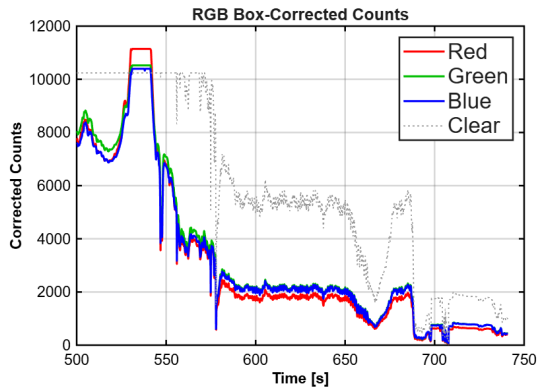


Figure 18: RGB counts vs Time

Figure 18 demonstrates that as the dive progresses and depth increases, all channels decrease in counts. As seen in Figure 17, Spectral Transmission  $T$  was analyzed and computed at each depth using the equation:

$$T = \text{counts}(\text{depth}) / \text{counts}(\text{surface}) \quad (8)$$

Where the surface reference just below the water is taken ( $I_0$ ) in Beer-Lambert.

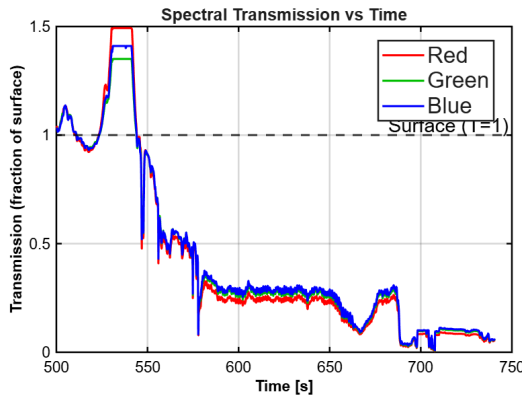


Figure 19: Spectral Transmission vs. Time

To compute for  $I_0$ , the period of time when the robot was stationary in water just before the robot descended (480-500 s) was found and the counts were used as reference to the surface. The reference was then used to compute for Spectral Transmission and Light Attenuation, as seen in Figures 19 and 20, respectively.

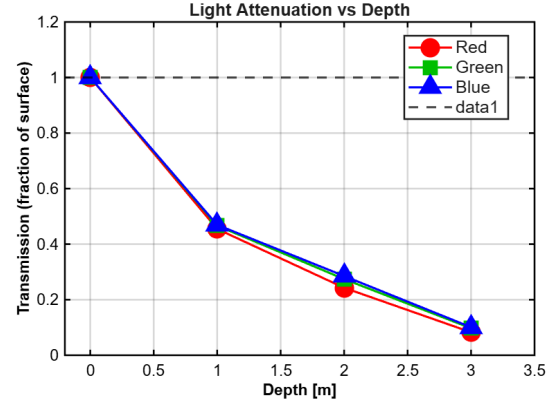


Figure 20: Light Attenuation vs. Depth in reference to the surface

Both Figures 19 and 20 are essentially demonstrating the same results, with the main difference being that Figure 20 is Light Attenuation as a function of depth (m). From Figure 19, Spectral Transmission falls at a rapid rate for all channels.

Figure 20 demonstrates similar results as a function of depth, showing median transmission at each tether waypoint (0, 1, 2, 3) m for RGB channels. The results are seen in Table 2.

| Depth (m)   | T_Red | T_Green | T_Blue |
|-------------|-------|---------|--------|
| 0 (surface) | 1.000 | 1.000   | 1.000  |
| 1.0         | 0.455 | 0.467   | 0.469  |
| 2.0         | 0.243 | 0.274   | 0.285  |
| 3.0         | 0.083 | 0.097   | 0.101  |

Table 2. Spectral Transmission at each Depth Waypoint (Dive 3)

At 3m, red wavelengths maintain only 8.3% of surface light intensity, green sees 9.7% and blue retains 10.1%. Throughout the dive, they follow the same pattern, consistent with Beer-Lambert predictions and the known seawater absorption spectrum [4]. By 3m, as seen in Figure 20, less than 20% of surface light was seen across all wavelengths. The small difference between green and blue transmission (0.4%) is consistent with what is seen in coastal waters (Figure 1). It is expected that at greater depths, there would be more pronounced spectral separation between the two channels.

As previously mentioned, kelp photosynthesis requires only 1% of surface PAR to function [6]. From the results, all color channels retain around 10% of surface light at 3m., thus photosynthesis is achievable

within the dive range. The maximum depth kelp can grow was not reached, but results suggest kelp can survive at depths greater than 3 m at Dana Point. Deeper dives would determine the exact range kelp ecosystems can thrive.

#### IV. CONCLUSION

##### *A. Overall Results*

The intended goal of the project was not fully achieved. However, the team accomplished meaningful outcomes. The RGB color sensor was able to successfully characterize light attenuation with depth at Dana Point, with results consistent with the predictions from known water light absorption behavior and Beer-Lambert law. As previously stated, red wavelengths retained only 8.3% of surface light intensity at 3 m, while green saw 9.7% and blue retained 10.1%. More than 80% of surface light was lost by 3 m.

The pressure sensor successfully performed as intended throughout the dive, reaching the final waypoint of 3 m, although a 0.5 m offset was corrected during post-processing. P-control allowed for the robot to descend to its final target depth, although the robot did not stop at the intended intermediate waypoints, likely due to a rapid descent.

The thermistor recorded a fixed temperature recording of 23.7°C throughout the entire dive, likely due to a mistake in waterproofing or malfunction in the thermistor itself. The circuit design had been validated, as well as bench testing to confirm the design functioned as intended.

During Dive 2, saltwater flooded the main electronics box, preventing further collection of pressure and temperature data and preventing the planned 6 m dive with multiple deployments at different times of day.

Note that staff and proctors recommended leaving the RGB dive until after collecting some data with the other sensors, due to the constant short-circuiting during the integration process. With the motherboard still functioning, Dive 3 was manually conducted for RGB data collection.

##### *B. Significance of Results*

The greatest finding of this project is that all three light channels, Red, Green, and Blue, lost greater than 80% of surface light intensity by 3 m depth, but remain above the 1% PAR required for kelp photosynthesis. This suggests that kelp can survive at depths greater than 3m, as the minimum threshold was not reached, requiring deeper dives to determine what depth range kelp can grow. Reaching the depth at which photosynthesis is at the threshold would require

additional data at deeper depths, such as the original intended depth of 6m, where stronger separation between channels as well as less light transmission would be expected. Our findings are ecologically significant, and kelp growth is also not limited to available light intensity, but other stresses such as temperature may be impacting the survival of kelp [11].

From an engineering perspective, the project highlights an important lesson of verifying component functionality before deployment as essential through the malfunctioning of the thermistor 1. Although a circuit design can be correct and completely functional, a faulty component heavily impacts the experimenting process. Additionally, it is essential to double check waterproofing and security of the electronics before deployment. On an economical standpoint, the RGB sensor was a low cost (\$7.90) sensor and demonstrated that it can collect meaningful attenuation data underwater after calibration.

##### *C. Future Work*

Future iterations of this project would be essential in order to meet the original intended goals of this project, with several improvements in mind. One of the essential improvements is to verify the functionality of the thermistor at known temperatures before deployment, as well as having backup waterproofed thermistors to prevent future malfunctioning incidents.

Additionally, water intrusion must be prevented. During the design process, waterproofing was successful; and tested at higher pressures. However, it is essential that during deployment at the kayak, multiple members of the team double check that no loose wires are preventing the full closure of the box. It is likely that during the second dive a loose wire led to the flooding of the main box, since during both dive 1 and 3, no flooding occurred. Designing a fixed wire route (tape) that prevents loose wires from obstructing the lid closing will improve the deployment of the UAV.

Given more time, completing the original plan of diving 6 m would provide more information regarding light attenuation and allow the identification of the depth threshold for kelp photosynthesis. Furthermore, deployments at multiple times a day would also allow analysis for the effect of solar angle on underwater spectral light, with information at what time of day is best. With multiple dives, less uncertainty can be concluded from the data.

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